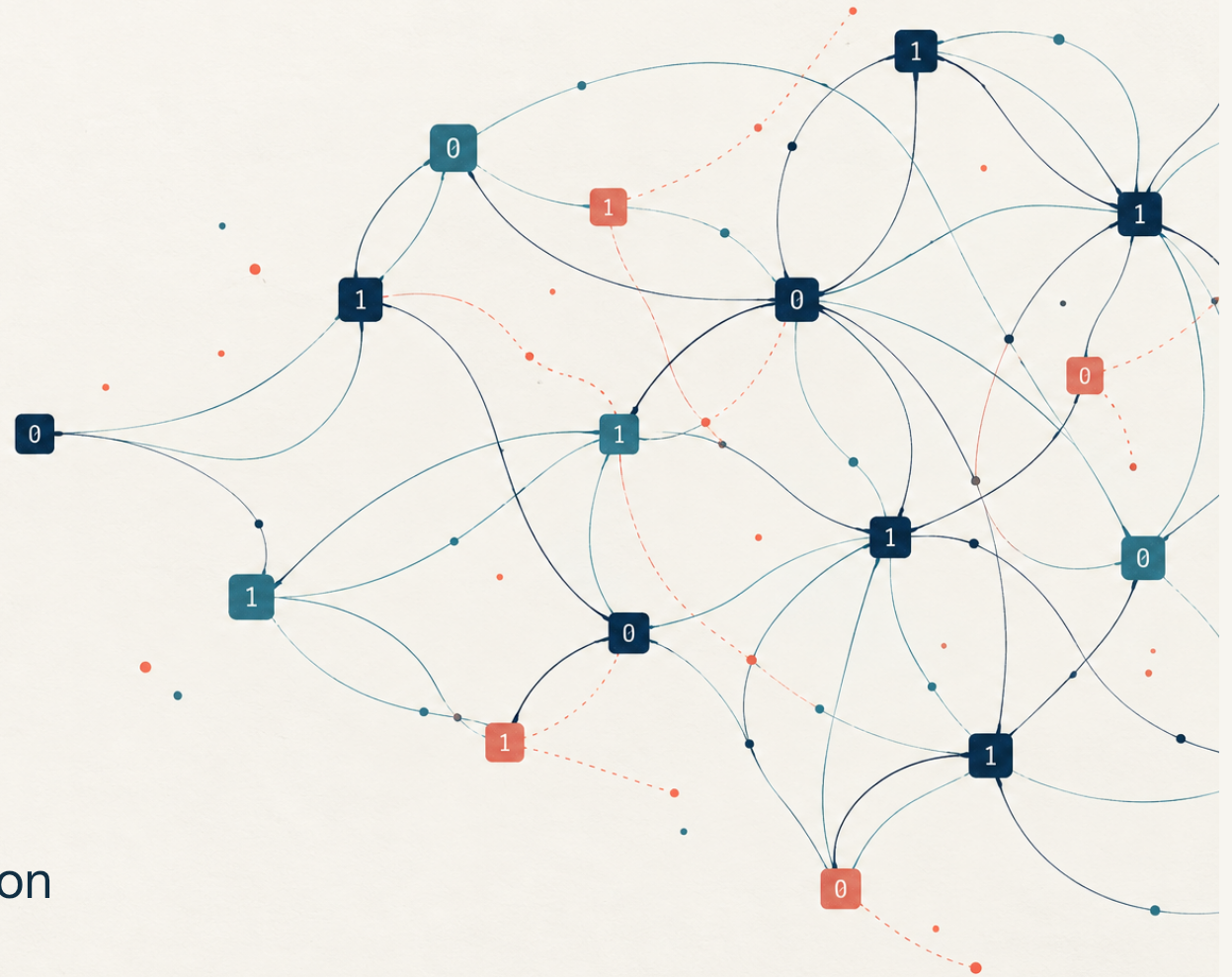


d-LIN

Sparse noisy linear systems

EXPERT SEMINAR

Definitions, assumptions and the ABW connection



The sampled d-LIN distribution

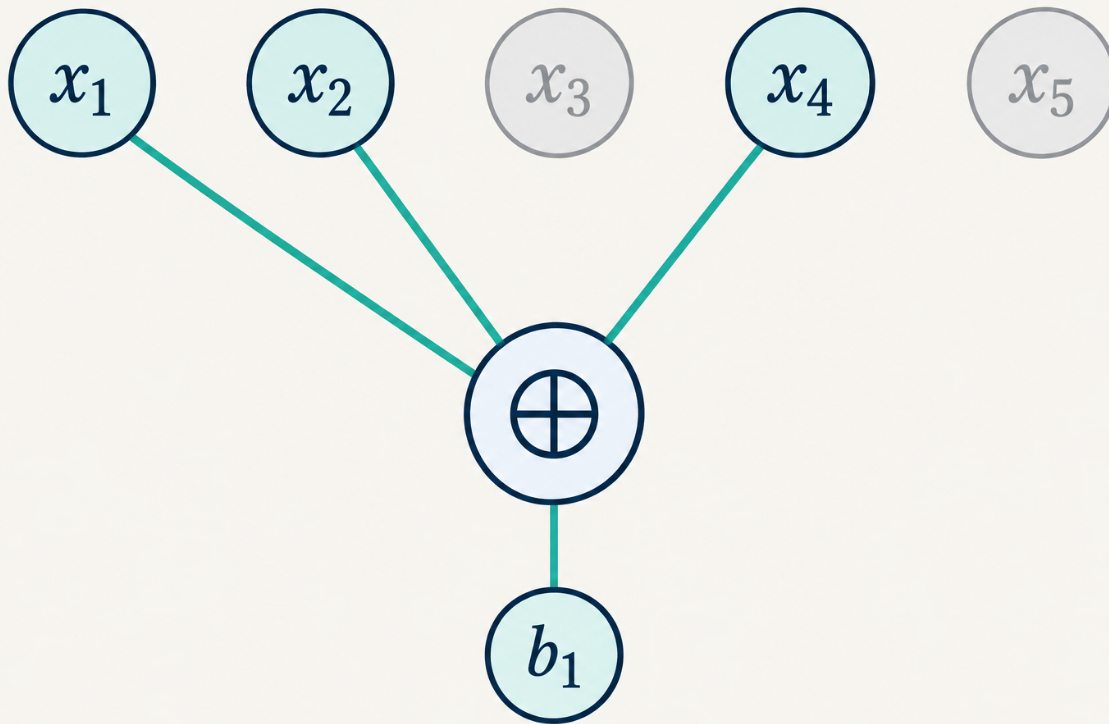
$$\mathbf{M} \leftarrow \mathbf{M}(m, n, d) \quad \mathbf{x} \leftarrow \mathbf{U}(\mathbb{F}_2^n) \quad \mathbf{e} \leftarrow \text{Ber}(\epsilon)^m$$

$$(\mathbf{M}, \mathbf{b}) = (\mathbf{M}, \mathbf{M}\mathbf{x} + \mathbf{e})$$

Parameter	Role
m / n	Equation count relative to variable count
d	Exactly d ones in each matrix row
ϵ	Independent output-flip probability

All arithmetic in the sampled system is over $\mathbb{F}_2$.

Sparse rows encode local constraints



$$\mathbf{b}_i = \bigoplus \mathbf{x}_j + \mathbf{e}_i$$

j in the support of row i

$$|\text{supp}(\mathbf{M}_i)| = d$$

Column weights can vary.

The row ensemble matters.

The diagram shows a noiseless row. Add $\mathbf{e}_i$ to model its observed output.

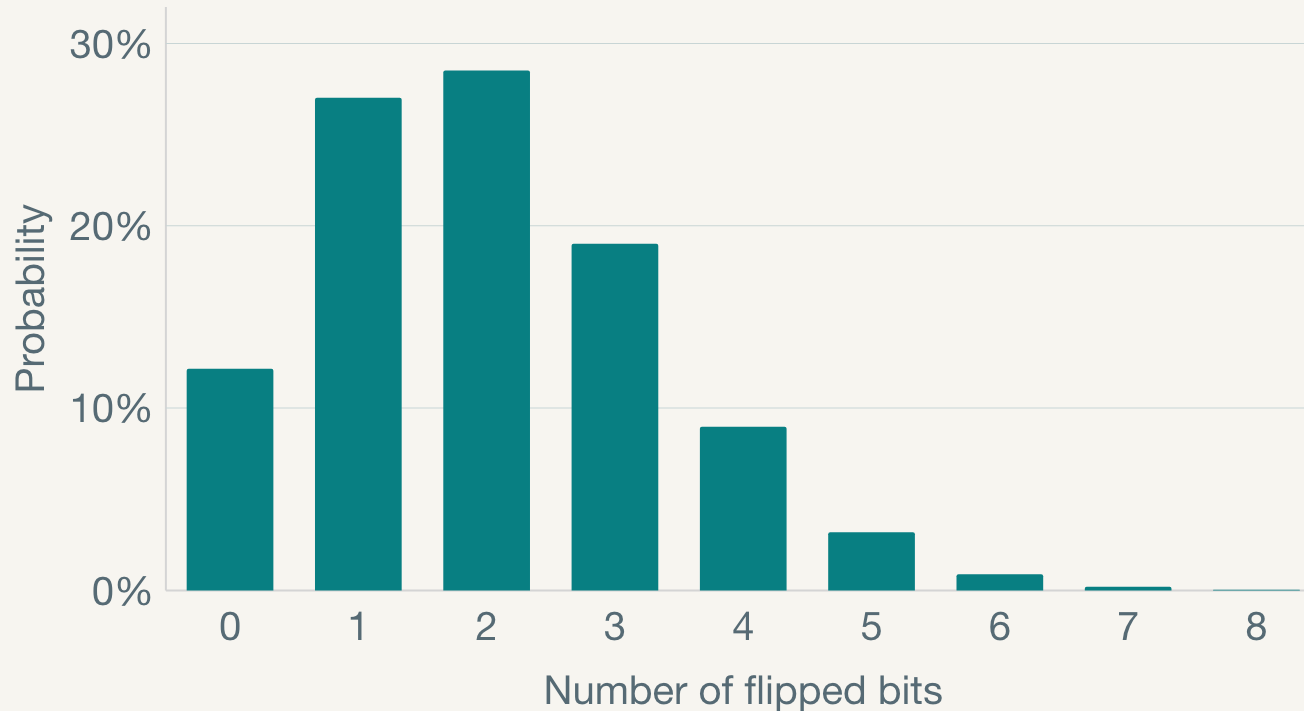
ABW task definitions and a separate optimization

Task	Target
Search3LIN	Recover the sampled x , with success at least $2/3$
AppSearch3LIN	Return $\hat{x}$ with $d_H(\hat{x}, x) \leq 0.1n$
Decision	Distinguish $(M, Mx + e)$ from (M, u) , u uniform
Residual optimization	Minimize $wt(Mz - b)$ over z

For even d , $M1 = 0$, so x and $x + 1$ produce identical clean labels.

Error count: expectation and concentration

Illustration: $m = 20$, $\epsilon = 0.10$



$W \sim \text{Binomial}(m, \epsilon)$

$$E[W] = m\epsilon$$

$$\text{Var}(W) = m\epsilon(1 - \epsilon)$$

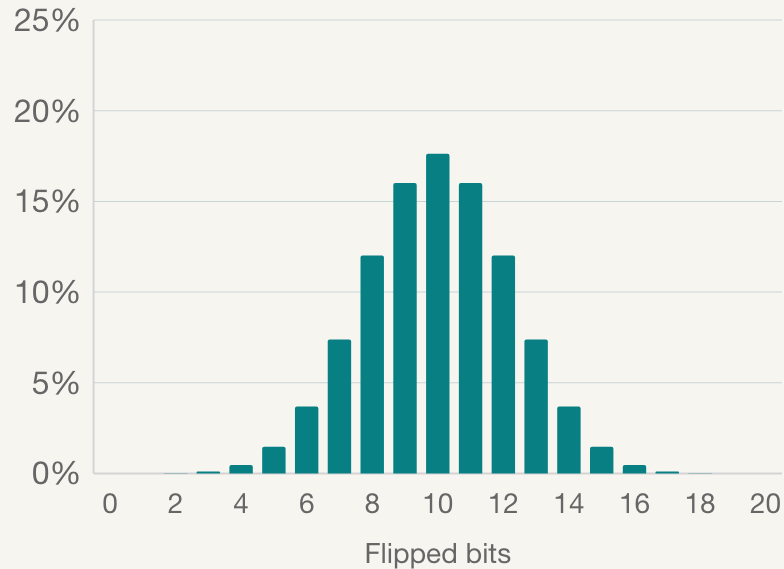
$$\Pr[W/m > \epsilon + t] \leq \exp(-2mt^2) \quad (t > 0)$$

A high-probability bound needs slack above ϵ , or a separate tail calculation.

The same error model at three noise rates

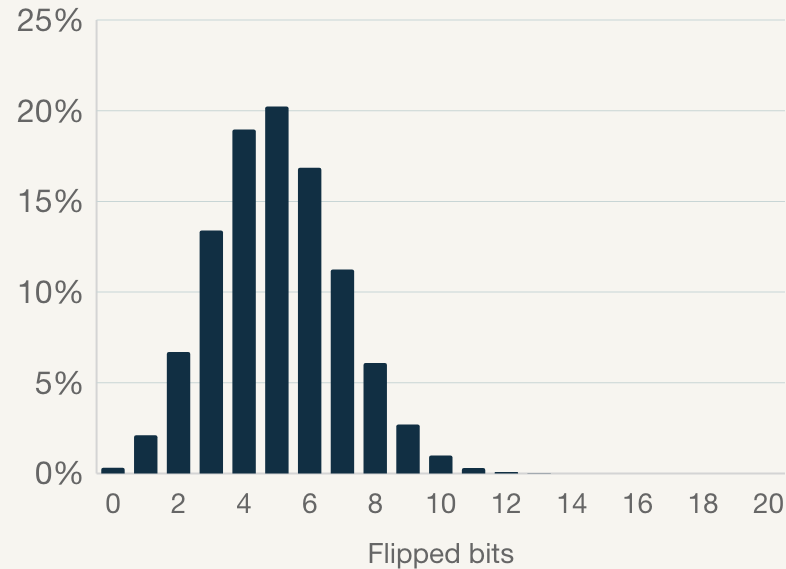
$W \sim \text{Binomial}(20, \epsilon)$

$\epsilon = 0.5$



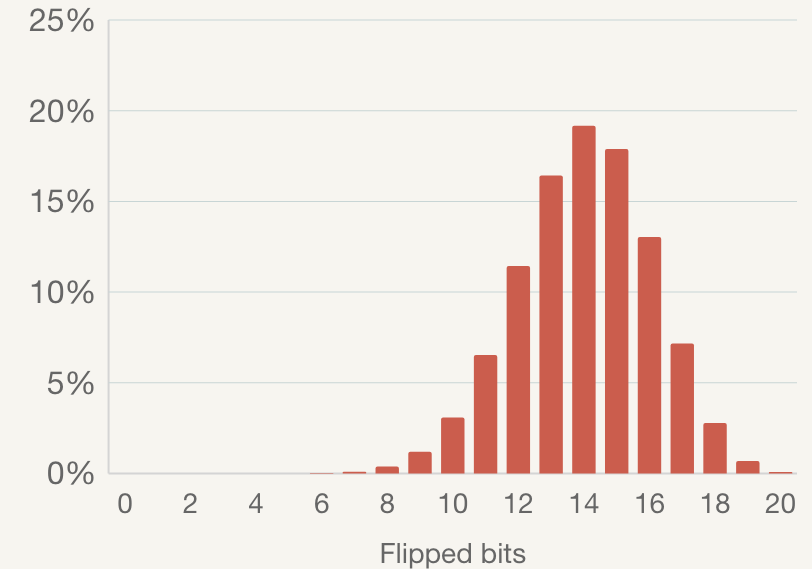
Mean = 10

$\epsilon = 0.25$



Mean = 5

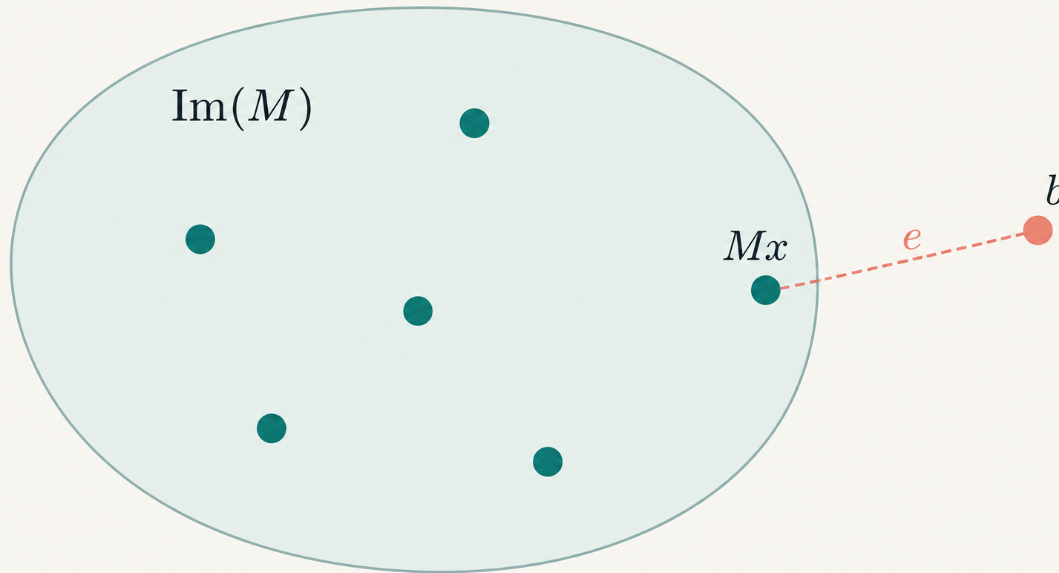
$\epsilon = 0.7$



Mean = 14

$\epsilon = 0.5$ hides x completely. Flipping all labels changes $\epsilon = 0.7$ into $\epsilon = 0.3$.

Distance to the planted output and to the image



$$C = \text{Im}(M)$$

$$dH(b, Mx) = \text{wt}(e)$$

$$dH(b, C) \leq \text{wt}(e)$$

$$E[dH(b, C)] \leq m\varepsilon$$

Equality to the planted distance does not imply equality to the nearest-image distance.

The notes' toy instance has an exact fit

M

1	1	0	1	0
0	1	1	0	1
1	0	1	1	0
0	1	0	1	1

$$\mathbf{b} = (1, 1, 0, 1)^\top$$

Sampled $\mathbf{x} = (1, 0, 1, 0, 1)^\top$

Residual weight = 1

$$\mathbf{x}' = (0, 1, 0, 0, 0)^\top$$

$$\mathbf{M}\mathbf{x}' = \mathbf{b}$$

$\text{rank}(\mathbf{M}) = 4$, so $\text{Im}(\mathbf{M}) = \mathbb{F}_2^4$. A planted error need not create distance from the image.

Hardness depends on the ensemble and task

Specification	Why it changes the claim
Matrix distribution	Row supports, dependencies and rank affect information
Parameters m, n, d, ϵ	Sample density and noise affect attack costs
Success criterion	Exact recovery, approximation and advantage differ
Adversary model	PPT, restricted circuit classes and other models differ

2ⁿ candidate assignments $\neq$ a hardness proof

Theorem 2: PKE from 3-LIN alone

$$3\text{LIN}(c n^{1.4}, n^{-0.2} / c)$$

For a suitable constant c , this assumption implies semantically secure public-key encryption.

Parameter	Asymptotic role
$m = \Theta(n^{1.4})$	Superlinear number of sparse equations
$\varepsilon = \Theta(n^{-0.2})$	Noise decreases, expected count is $\Theta(n^{1.2})$
DUE	No DUE assumption in this theorem

Theorem 3: d-LIN together with DUE

$\text{dLIN}(c' n \log n, 1 / (c' q))$

$\text{DUE}(cn, q, 2d)$

Both assumptions imply semantically secure PKE for the theorem's parameter regime.

DUE hides a set of q rows supported on only $q/3$ columns. The public matrix has locality $2d$.

Here q counts planted rows. It is unrelated to the modulus q used in LWE.

d-LIN, LPN, LWE and Module-LWE

Problem	Linear structure	Error model
d-LIN	Binary rows with exactly d ones	Independent bit flips
LPN	Uniform binary rows, typically dense	Independent bit flips
LWE	Uniform matrices over $\mathbb{Z}_q$	Small centered modular errors
Module-LWE	Polynomial matrices over R_q	Small polynomial coefficients

Changing the matrix distribution or the noise changes the assumption.

LWE and Module-LWE structure

$$\text{LWE: } \mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e} \pmod{q}$$

A contains ordinary modular coefficients. Small errors perturb the outputs.

$$\text{Module-LWE: } \mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e} \text{ in } Rq^k$$

A, s and e contain polynomials. Coefficient expansion creates correlated convolution blocks.

Module structure is different from fixing d nonzero bits in each d-LIN row.

What hardness support each setting provides

Setting	Relevant support and its limits
d-LIN / ABW	Average-case assumption, task reductions and restricted attack evidence
LPN	Separate binary-noise assumptions. PKE depends on parameters and quantitative hardness
LWE	Worst-case lattice reductions under specified parameters and error laws
Module-LWE	Reductions from module-lattice problems under specified conditions

The original LWE and Module-LWE reductions cited here are quantum reductions.

A short row dependency removes x

Four 3-sparse rows

1	1	1	0	0	0
1	0	0	1	1	0
0	1	0	1	0	1
0	0	1	0	1	1

$$\sum_{i \in S} M_i = 0$$

$$\sum_{i \in S} b_i = \sum_{i \in S} e_i$$

The hidden assignment cancels
when the selected rows sum to zero.

Each column appears twice in this illustrative dependency.

The ABW bit-encryption mechanism

Public key	Secret key
Sparse matrix M	Short set S with $\sum_{i \in S} M_i = 0$ and $m \in S$

$$\text{Enc}(\sigma) = Mx + e + \sigma u_m$$

$$\text{Dec}(c) = \sum_{i \in S} c_i = \sigma + \sum_{i \in S} e_i$$

σ is the message bit. u_m has a 1 only in the last position.
Fresh x and e randomize every encryption.

Key generation must hide the short dependency

Theorem 2 route

Condition an appropriate random 3-sparse ensemble on a short dependency.

The proof establishes sufficient statistical overlap.

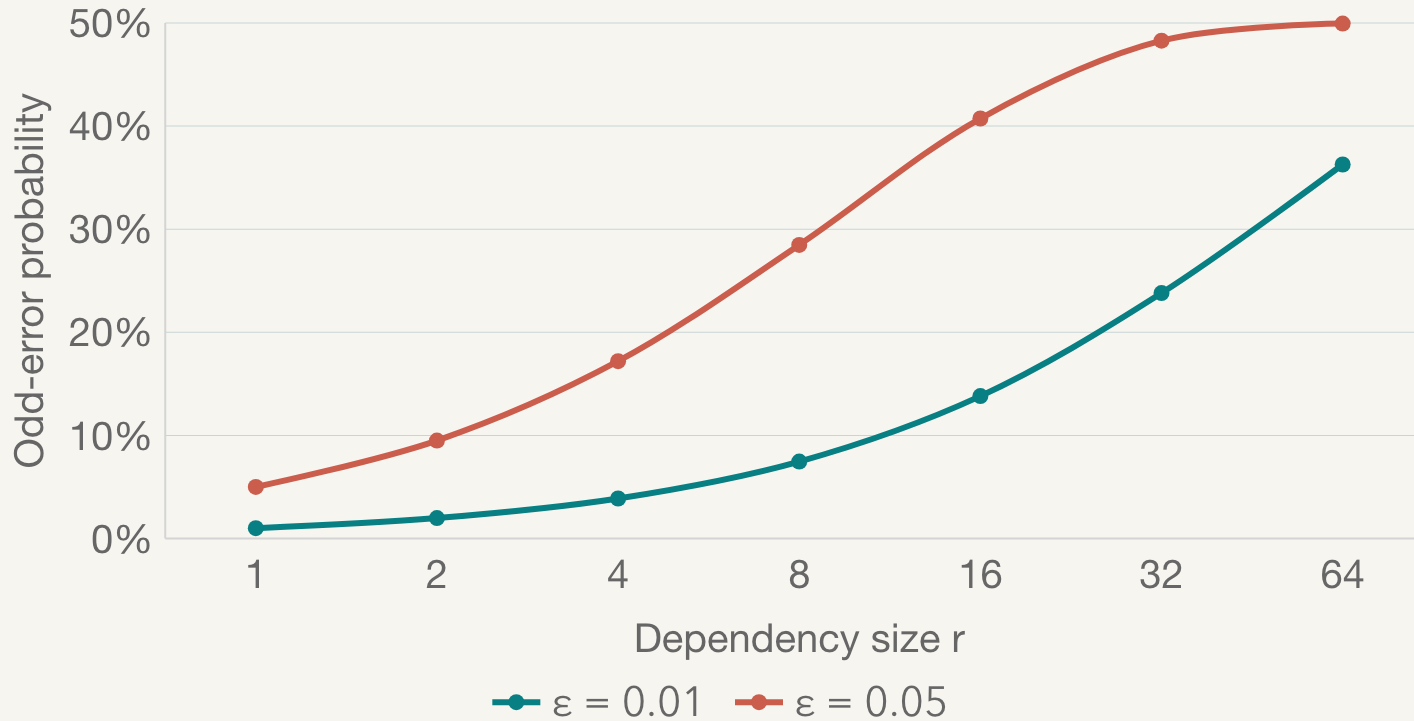
Theorem 3 route

Plant a shrinking set of rows with too few neighbors.

DUE controls whether an observer can detect this structure.

Finding an arbitrary left-kernel vector does not implement the required short trapdoor.

Noise accumulates along the dependency



$$p = [1 - (1 - 2\epsilon)^r] / 2$$

$$p \leq r\epsilon$$

Short dependencies preserve a useful parity bias.

The horizontal labels double r at each step. Values follow the independent-noise model.

Weak encryption and amplification

$$\alpha < (1 - \sqrt{\beta}) / 2$$

ABW first obtains constant correctness and privacy. Amplification gives semantic security and reliable encryption.

Quantity	Meaning
α	Raw decryption-error bound
β	ABW privacy parameter
Section 7 noise conversion	$\epsilon_{\text{cipher}} = 2\eta(1 - \eta)$, where η is search noise

Evidence covers particular attack families

Family	Scope of the evidence
Myopic algorithms	Restricted access to noisy outputs or their combinations
Low-degree polynomials / AC^0	Restricted algebraic or circuit computation
Lasserre / SDP hierarchy	Specified relaxation levels and parameter regimes

A cryptographic assumption remains after these restricted lower bounds.

Reasons to study d-LIN for public keys

A different foundation

ABW connects natural sparse noisy constraints to public-key encryption.

Simple local arithmetic

Core evaluation uses $O(md)$ bit operations. Row index lists use $O(md \log n)$ bits.

A proved conditional construction

Theorems turn specified average-case hardness assumptions into semantic security.

Costs and unresolved issues for public keys

Security assumptions

Parameters and sampling matter.

A short hidden dependency must remain hard to exploit.

Restricted attack evidence leaves a general hardness assumption.

Implementation costs

Noise and dependency length compete with correctness.

Amplification adds overhead.

Concrete parameters, CCA security and quantum analysis need more work.

Cheap sparse XOR operations alone do not establish an efficient final cryptosystem.

ABW and a concrete Module-LWE standard

Aspect	ABW d-LIN construction	ML-KEM
Role	Asymptotic PKE existence result	Standardized key encapsulation
Security statement	Semantic security under stated assumptions	IND-CCA2 goal under Module-LWE and proof conditions
Parameters	No concrete 128-bit set in ABW09	ML-KEM-512 / 768 / 1024
Example sizes	No like-for-like benchmark here	ML-KEM-768: key 1184 B, ciphertext 1088 B

PKE encrypts messages. A KEM establishes a shared secret for a larger protocol.

Reading path and main distinctions

Question	Key distinction
What is the task?	Exact recovery, approximate recovery, decision or residual fitting
What makes it hard?	A precise sampled distribution and an unproved average-case assumption
What does ABW prove?	Semantic-security existence under specific theorem hypotheses
How does it compare?	Same noisy-linear template, different distributions and reductions

ABW09: §§2, 5–7 and 9

Regev on LWE, Langlois–Stehlé on Module-LWE, NIST FIPS 203

Technical definitions and source links appear in the speaker notes.